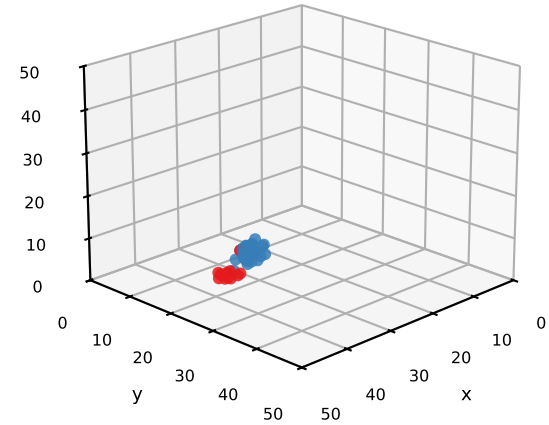
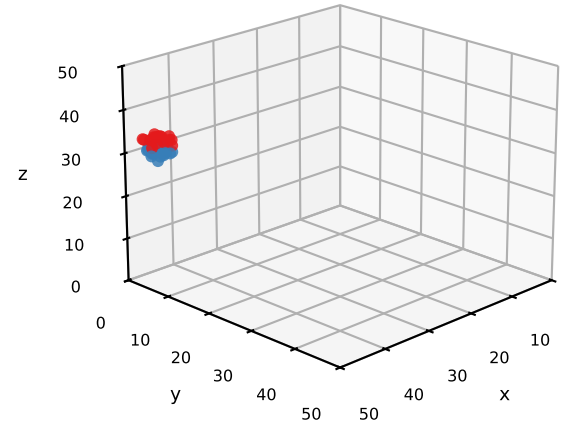


3D scenario snapshots — Fiedler partition coloring (red/blue = Fiedler bipartition; leadership $\lambda=2.4$ leaders shown larger)

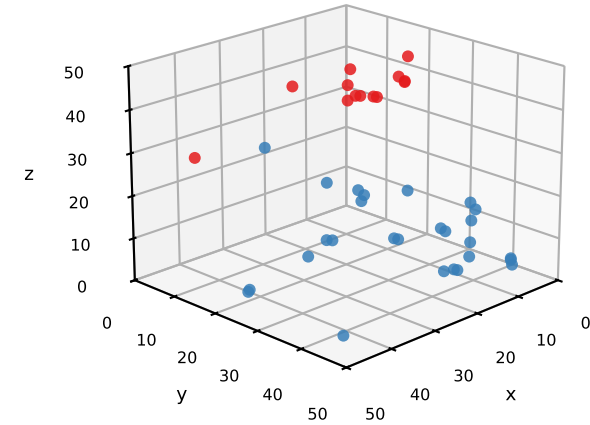
none / vanilla
 $\Phi=135.0$, $t=425$
(Fiedler partition)



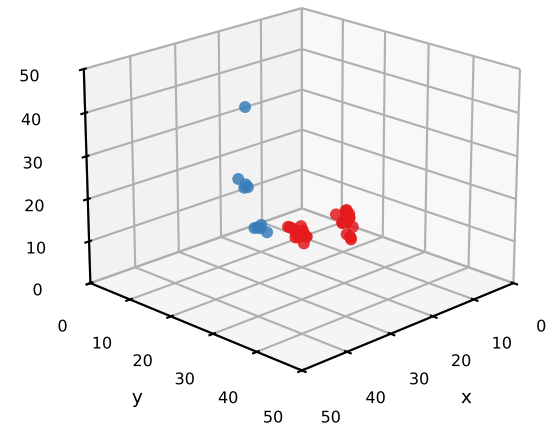
alignment $w_a=0.6$
(§4.2)
 $\Phi=173.4$, $t=320$
(k-means proxy)



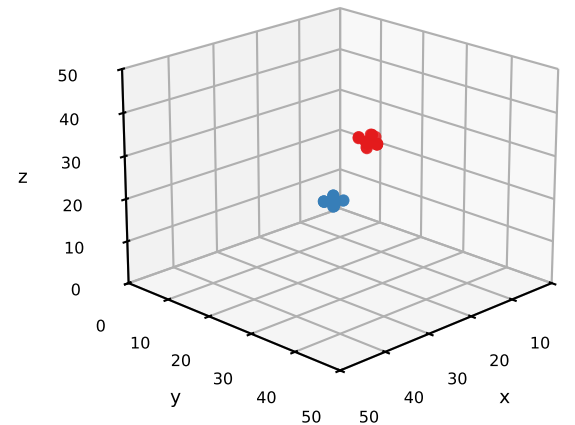
jamming $\alpha=0.2$
(§4.3)
 $\Phi=186.6$, $t=290$
(k-means proxy)



split_merge
(exploratory)
 $\Phi=168.2$, $t=230$
(k-means proxy)



milling $\mu=0.8$
 $\Phi=679.3$, $t=455$
(k-means proxy)



leadership $\lambda=2.4$
(leader-block)
 $\Phi=8.7$, $t=445$
(Fiedler partition)

